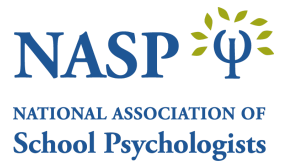


Communiqué

RESEARCH-BASED PRACTICE

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The World Federation of ADHD International Consensus Statement: Implications for School Psychologists

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School psychologists can play an important advocacy role in ensuring the latest evidence-based practices for students presenting with symptoms of attention deficit hyperactivity disorder (ADHD) are being implemented in their districts (DuPaul & Jimerson, 2014). ADHD is a highly stigmatized disorder due to a number of misconceptions (e.g., the prevalence of ADHD has increased in the last 20 years, ADHD is more common in the United States than other countries, ADHD often goes away or if left untreated will have a limited impact on one's future life functioning, psychostimulant medication only leads to short-term improvements) that persist among the general population (Faraone et al., 2021, p. 792). For example, a cross sectional study of TikTok videos found 52% of videos about ADHD contained misleading information, demonstrating how misinformation might be spread (Yeung et al., 2022). When combined with a negative stigma, a lack of knowledge about the etiology and developmental pathways surrounding ADHD can create significant disparities, barriers, and delays in providing effective educational services to youth at risk for or diagnosed with ADHD.

To promote a more accurate understanding of ADHD, The World Federation of ADHD published a consensus statement (Faraone et al., 2021) that comprehensively summarized the results of the latest scientific evidence about the developmental trajectory, outcomes, and treatments for ADHD. A group of 80 internationally distinguished ADHD researchers investigated 229 studies involving (a) 2,000 or more participants or (b) meta-analytic studies involving five or more studies (or had >2,000 participants). From this extensive review, 208 empirically supported statements were definitively concluded from the existing research literature and the following 14 summary statements were highlighted:

- ADHD has been described in the literature for almost 250 years.
- The diagnosis of ADHD is well-defined and valid for all ages.
- ADHD is more common in males and occurs in 5.9% of youth.
- ADHD is caused by the combined effects of genetic/environmental factors.
- Psychological tests of brain functioning cannot be used to diagnose ADHD.
- Neuroimaging procedures cannot be used to diagnose ADHD.
- People with ADHD are at increased risk for numerous health conditions.
- People with ADHD are at increased risk for poor developmental outcomes.
- The economic burden of ADHD to society is estimated in the hundreds of billions of dollars.
- Several medications are safe and effective in reducing ADHD symptoms.

- ADHD medications can reduce risky behavior (e.g., substance use, accidents).
- ADHD medication side effects are typically mild.
- Stimulants are more effective than nonstimulants, but more frequently diverted.
- Nonmedication treatments are less effective than medication, though essential.

The purpose of this article was to closely examine the 208 empirically supported ADHD consensus statements (Faraone et al., 2021) with an eye toward implications for the roles and functions of school psychologists. We do so within three primary job responsibilities, including screening/identification, mitigating developmental risk, and best practices in school-based interventions. We highlight how the latest evidence base surrounding ADHD might improve school-based mental health services for school-aged children at-risk for or diagnosed with ADHD. In Table 1, we highlight the specific consensus statements that we believe have the most significant implications on the practice of school psychology.

Table 1. Recommendations for School Psychologists’ Roles and Responsibilities

MAIN FINDING	BROAD IMPLICATIONS FOR SCHOOL PSYCHOLOGISTS	SPECIFIC ACTION ITEMS
Screening and Identification		
• Increased assessment for diverse populations. • ADHD impairs functioning across low, average, and high IQ, but can be masked by higher IQ. • Increased monitoring of symptoms, they change over time but continue to affect students as they age, and look different in diverse populations. • Presentation of ADHD predicts different impairment; inattention predicts academic impairment, low self-esteem, negative occupational outcomes and lower adaptive functioning. Hyperactive/impulsive symptoms are associated with peer rejection, aggression, risky behaviors, and accidental injuries.	• School psychologists should provide psychoeducation to dispel myths and provide information for the public (form of universal prevention). • School psychologists should also ensure we are providing ecologically valid diagnoses of ADHD.	• Psychoeducation can be published via school newsletters and information sessions. • School psychologists can consult literature, colleagues, and other healthcare providers to ensure diagnoses are ecologically valid. These diagnoses should predict real-world behavior. • School psychologists should stay up to date with the current research to best screen for and identify students with ADHD symptoms. • Evaluations of cognitive abilities including executive functioning can also be conducted early on.
Mitigating Developmental Risks: Focus on Prevention and Improving Outcomes		

MAIN FINDING	BROAD IMPLICATIONS FOR SCHOOL PSYCHOLOGISTS	SPECIFIC ACTION ITEMS
Environmental risk factors prior to diagnosis (deprivation, stress, infection, poverty, and trauma). Risk after diagnosis: physical and mental health issues, legal issues, educational underachievement, and the economic burden of ADHD (it costs significantly more to raise a child with ADHD).	- School psychologists should work toward improving outcomes by spreading knowledge and advocating for systems-level change. - School psychologists should inform others about the developmental risks associated with ADHD and the deleterious outcomes that may arise from use of ineffective treatments.	School psychologists can provide information within their schools and communities by spreading awareness and knowledge to help families with children with ADHD (e.g., holding general psychoeducation sessions for schools and communities). School psychologists can advocate with teachers, parents, and physicians for systems-level change to provide more support for children and families through community resources.

Best Practices in ADHD Interventions

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| <ul style="list-style-type: none"> Parent or teacher training. Cognitive training (preschoolers). CBT (older students). Organizational skills. Medication. | <ul style="list-style-type: none"> School psychologists should choose the most appropriate treatment, which is mainly determined by age (however, this could also vary by symptomology). School psychologists must be informed and share with others those ADHD treatments that have moderate to strong research support. | <ul style="list-style-type: none"> Parent training is most effective for young children (can be done with parent-training groups). Cognitive training is effective for preschoolers and can be used to reduce off-task behavior and enhance social skills. CBT is more effective for older children and adolescents who can engage in cognitive restructuring and participate in their own behavior plans. CBT can be used with individuals and in group settings (e.g., groups in high school settings). Organizational training can be used to alleviate symptoms of inattention. School-based medication monitoring can help to ensure maximum benefits while limiting costs (i.e., side effects). |
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Screening and Identification of ADHD

The meta-analyses the World Federation examined found that there has not been an increase in the overall prevalence of ADHD in the past 3 decades; however, ADHD is more likely to be diagnosed in children and adolescents due to improved identification of the disorder, as well as changes in administrative and clinical practice. ADHD is more common among males, indicating an actual sex difference, and among Black individuals. The World Federation's statement called for increased screening, assessment, and monitoring of ADHD among diverse populations, particularly among Black individuals. These findings indicate a need for school psychologists to screen, assess, and monitor ADHD in diverse school-age populations. This includes the need for assessment of inattention, hyperactivity, and impulsivity in students from preschool to high school. In addition to accounting for developmental and age differences within assessment practices, school psychologists also need to closely examine other demographic characteristics linked to ADHD, including race, ethnicity, sex, and socioeconomic status. This is especially important given that ADHD research has historically focused on White boys from upper-middle-class families. Improved school-based behavioral health screening and assessment practices that are individualized to the presenting student population, the school context, and the community in which the school is located are especially important to implement.

To aid in the improved diagnosis and identification of ADHD, the World Federation reviewed the current practices for diagnosing ADHD, outlining the critical components of an evidence-based diagnosis. Essential for a diagnosis of ADHD is the presence of ADHD symptoms (inattentive or hyperactive/impulsive) for at least 6 months, with symptoms occurring in two or more settings that cause impairment in different areas of daily functioning. Symptoms must have begun in early childhood and cannot be better explained by the presence of another psychological disorder or physical illness. While diagnoses of ADHD have been criticized as being subjective, the World Federation found that the well-defined criteria and the guidelines of using evidence-based, multimethod assessments (e.g., diagnostic interviews and rating scales) rebuke this criticism. Additionally, while there are structural and functional differences in the brains of individuals with ADHD, those differences are not sensitive enough to be diagnostically relevant with current technologies. School psychologists are well trained in assessment and have the opportunity to emphasize ecologically valid ADHD screening and identification practices. They also are well positioned to provide psychoeducation within their districts about the validity of ADHD diagnoses, implement evidence-based ADHD assessment approaches, and address harmful misconceptions of ADHD among the public. By engaging in differential diagnostic processes, school psychologists can also help others to recognize that ADHD often co-occurs with other disorders (e.g., depression, anxiety, autism spectrum disorders) and that comprehensive early identification and intervention within school settings is paramount to improving student outcomes.

Mitigating Developmental Risk: Focus on Prevention and Improving Outcomes

The Federation also reviewed meta-analyses regarding the cause of ADHD. Many different genetic and environmental risk factors compound to lead to both the onset of the disorder and its impact on functioning. Understanding the genetic factors that lead to ADHD and their interaction with environmental factors is essential for school psychologists in terms of providing psychoeducation to students, families, teachers, and other school personnel. The strong associations between several environmental etiologies linked to ADHD additionally indicate other roles for school psychologists to play. These include advocacy or other efforts to eliminate exposure to harmful environments (e.g., lead exposure, prenatal exposure to nicotine) that are known to increase the risk of ADHD. Public health efforts to reduce prenatal and postnatal nutrient deficiencies (e.g., omega-3 PUFAs and vitamin D) is another role. Finally, working closely with expectant mothers to address factors associated with low birth weight and birth trauma may also be impactful in preventing ADHD. Additional environmental risk factors for ADHD to be targeted for prevention and early intervention include stress, infection,

poverty, and trauma. These postbirth findings demonstrating considerable inequities provide a call to action for school psychologists in the form of social justice. Advocating for increased equity in social services, housing, healthcare, and access to evidence-based interventions can help directly mitigate factors linked to ADHD.

The World Federation also examined several negative developmental outcomes associated with ADHD including physical health issues, lower quality of life (for individuals with ADHD and their families), and economic impacts. In terms of poor health outcomes, individuals with ADHD are more likely to experience higher rates of obesity, asthma and allergies, and diabetes (specifically type 2 diabetes). In addition, ADHD has been linked to sleep issues, psoriasis, autoimmune disorders, epilepsy, sexually transmitted infections, kidney failure, and high blood pressure. School psychologists can work to screen for and identify students experiencing ADHD symptoms and then provide targeted health interventions to reduce the functional impact of these developmental issues.

In addition to the health impact of ADHD, the Federation reports that deficits in social, emotional, and behavioral health are more prevalent in students with ADHD when compared to typically developing peers. Higher emotional reactivity and challenges in moderating those reactions is a common experience for students with ADHD. Additionally, students with ADHD may have problems with social skills, such as difficulties in accurately identifying and understanding the meaning of social cues. School psychologists can target emotional and social impairment for interventions (e.g., emotional regulation skills, social skills training). Youth with ADHD also experience higher rates of suicidal ideation and suicide attempts, which highlights the need to offer a range of supports for those with ADHD in an effort to prevent the emergence of co-occurring emotional difficulties like depression (e.g., hopelessness, helplessness) and anxiety (e.g., distorted cognitions, worry, fear).

The Federation's report also highlighted that youth with ADHD have been found to be more likely than their peers to engage in violent crimes, including higher rates of dating violence. They are also more likely to be convicted of crimes and subsequently incarcerated. Adolescents and young adults with ADHD are also more likely to make false confessions. Given the aim of social justice within school psychology, this once again provides the opportunity for school psychologists to look for ways to connect to their local juvenile justice court systems to explore ways to promote mental health diversion programs for those diagnosed with ADHD.

Educationally, the Federation's report concluded that individuals with ADHD are clearly more likely than their peers to have lower academic achievement and experience poor school outcomes. This includes not graduating from high school on time and dropping out. In addition, youth with ADHD exhibit poorer performance on overall expressive, receptive, and pragmatic language measures. School psychologists' involvement in academic and organizational skills interventions can help to address deficits to improve academic outcomes in those with ADHD.

Particular attention in the Federation's report is given to the link between ADHD and numerous other poor outcomes including alcohol, nicotine-related, and substance use disorders. ADHD also increases the risk of gambling, reckless spending, and impulsively quitting a job. In addition, girls with ADHD are more likely to become teenage mothers. Finally, the economic impact (e.g., lower lifetime earnings) associated with ADHD significantly impacts families and, more broadly, society as a whole. Reducing these additional risks and costs associated with ADHD is another role for school psychologists to embrace.

Best Practices in School-Based Interventions

There are many nonmedical treatments for ADHD; behavioral therapies for ADHD vary widely, mainly based on the patient's age. Parent training is offered for younger children to help parents interact with their children and manage their ADHD symptoms effectively. Organizational skills training is helpful for adolescents and young

adults to manage their daily life. Cognitive-behavioral treatment (CBT) is often provided in addition to organizational training. These treatments are associated with moderate improvement in functioning. Furthermore, there are varying levels of evidence for behavioral and cognitive therapies, cognitive training and neurofeedback, and nutrition and diet. School psychologists can provide many of these nonbiological treatments and collaborate with other healthcare professionals in school and healthcare settings to provide the other comprehensive treatments. Due to a number of developmental considerations within a cost-benefit analysis of effective treatments for ADHD, it is recommended to try these psychosocial treatments prior to using medication. Meditation-based therapies have not been empirically supported in reducing ADHD symptoms, and social skills training was not shown to improve social skills, general behavior, or school performance.

Medications are both safe and effective for ADHD. A substantial body of literature highlights the effectiveness of stimulants for reducing symptoms of ADHD (such as response inhibition). In addition to treating ADHD, stimulant medications have also been shown to improve functioning for individuals with co-occurring emotional dysregulation, borderline intellectual disabilities, anxiety, aggression, and oppositional behavior, including conduct disorder. Furthermore, ADHD medication reduced the likelihood of developing depression and suicidal thoughts and actions. Beyond psychological symptoms, medication for ADHD has been shown to improve academic performance, and reduce criminality and accidental injuries (including traumatic brain injuries and bone fractures). ADHD medication was also associated with reduced motor vehicle accidents for males. Nicotine dependency was reduced, and it has been found that stimulant medication does not increase the risk for dependence on alcohol or other substances (e.g., marijuana and cocaine). ADHD medication was associated with a decrease in teenage pregnancy as well. Overall, stimulants show a reduction in ADHD symptoms and the related risks outlined above. While school psychologists are not typically prescribers of ADHD medication, they can provide referrals and psychoeducation about stimulant medications and collaborate with other professionals to provide coordinated care. Some side effects of stimulant medications include sleep disruption, appetite decreases, some delays in height gains, and cardiovascular events (in rare cases). There is also a risk of misuse (e.g., taking too much or inconsistently with the doctor's prescription) and diversion (being taken by individuals not prescribed the medication), particularly among college students. Misuse and diversion are associated with lower educational attainment, and stimulants are not related to improved grades for individuals without ADHD. Indeed, school psychologists, particularly those working with college students and adolescents, may want to be wary of misuse and diversion.

Computer-based cognitive training and neurofeedback have become popularized on the internet, but have been shown to minimally reduce ADHD symptoms. However, they did lead to some short-term improvements in verbal working memory and visuospatial working memory. However, there was no evidence indicating that these effects would be long-lasting. As such, school psychologists may want to avoid recommending these, and instead focus on providing treatment options with more empirical support.

Finally, emerging literature has shown small effects of reducing ADHD symptoms when increasing the intake of omega-3 fatty acid, as well as when reducing synthetic dyes from children's diets. Exercise has been shown to primarily provide improvements in reducing comorbid symptoms of depression and anxiety. There is some evidence that healthier eating habits may reduce inattentive symptoms. Regarding nutrition, school psychologists may counsel students and families to eat healthily and exercise, while recognizing that behavioral treatments and psychostimulant medications are the first line of treatment for ADHD in school-age populations.

Conclusion

Overall, the World Federation of ADHD has reached a clear consensus that ADHD is a chronic disorder, in which inattention, hyperactivity, and impulsivity affect daily living in many areas, such as academic functioning, other mental health conditions, physical risk, and financial effects for individuals, families, and society. In addition, there is strong evidence that medication can help reduce most of the symptoms and risks associated with ADHD, and parent training and organizational skills training can supplement medication in order to improve the lives of individuals with ADHD. School psychologists have the opportunity to (a) carry out school-based psychosocial interventions within a progress monitoring framework, (b) support efforts to engage in the evaluation of the impact of adjunctive medication treatment on school performance (Carlson & Barterian, 2019), connect families to these services, and collaborate with other professionals to coordinate care. In addition to providing direct services, school psychologists can provide valid diagnoses, psychoeducation to individuals and their communities at large, and work to prevent some risk factors that lead to the onset of ADHD. While there are still gaps in the literature, such as the impact of treatments over time, the World Federation has provided outlines for practitioners in the field of school psychology to improve lives of those with ADHD.

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